

Test: VPC with NAT gateway

Step no:1 Create a VPC named: my-vpc-01, with an CIDR range of 10.0.0.0/16.

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

my-vpc-01

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

VPC encryption control (S) [Info](#)
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode ☐ Enforce mode

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q my-vpc-01 X [Remove tag](#)

[Add tag](#)

You can add 49 more tags

[Cancel](#) [Preview code](#) [Create VPC](#)

Step no:2 Create a public subnet named: my-public-subnet-1 for our VPC with a subnet range of: 10.0.1.0/24

Create subnet [Info](#)

VPC

VPC ID
Create subnets in this VPC.

vpc-00a1651d57d5137a6 (my-vpc-01)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

my-public-subnet-1
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24 256 IPs

Tags - optional

Key **Value - optional**

Q Name X Q my-public-subnet-1 X [Remove](#)

[Add new tag](#)

You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

Step no:3 Create a private subnet named: my-private-subnet-2, for our VPC with a subnet range of 10.0.2.0/24

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-00a1651d57d3137a6 (my-vpc-01)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

my-private-subnet-2

The name can be up to 256 characters long.

Availability Zone

[Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (Ohio) / use2-az1 (us-east-2a)

IPv4 VPC CIDR block

[Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24

256 IPs

Tags - optional

Key

Q Name

Value - optional

Q my-private-subnet-2

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

Step no:4 Create a routable for the public subnet named: my-public-route-table1 and mention the VPC as our my-vpc-01.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

my-public-route-table1

VPC

The VPC to use for this route table.

vpc-00a1651d57d3137a6 (my-vpc-01)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q my-public-route-table1

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Once created → go to subnet associations → choose our public subnet for associations → save changes.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR
☒	my-public-subnet-1	subnet-0713027daa63722df	10.0.1.0/24
☐	my-private-subnet-2	subnet-08015c59f2b49036d	10.0.2.0/24

Selected subnets

subnet-0713027daa63722df / my-public-subnet-1

Cancel Save associations

Step no:5 Create a routable for the public subnet named: my-private-route-table2 and mention the VPC as our my-vpc-01.

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

my-private-subnet-2

VPC
The VPC to use for this route table.

vpc-00a1651d57d3137a6 (my-vpc-01)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key Value - optional

Q Name X Q my-private-subnet-2 X Remove

Add new tag
You can add 49 more tags.

Cancel Create route table

Once created → go to subnet associations → choose our private subnet for associations → save changes

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☐	Name	Subnet ID	IPv4 CIDR
☐	my-public-subnet-1	subnet-0713027daa63722df	10.0.1.0/24
☒	my-private-subnet-2	subnet-08015c59f2b49036d	10.0.2.0/24

Selected subnets

subnet-08015c59f2b49036d / my-private-subnet-2

Cancel Save associations

Step no:6 Go to internet gateway → create an IGW named as: my-internet gateway.

Create internet gateway

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

my-internet-gateway

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key Value - optional

Q Name X Q my-internet-gateway X Remove

Add new tag
You can add 49 more tags.

Cancel Create internet gateway

Once created → actions → attach to VPC → choose our VPC → attach.

Attach to VPC (igw-03e28b8f88f44d5d0) [Info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

Q vpc-00a1651d57d3137a6 X

Use: "vpc-00a1651d57d3137a6"

vpc-00a1651d57d3137a6 - my-vpc-01

[Cancel](#) [Attach internet gateway](#)

Step no:7 Go to NAT gateway → create NAT gateway: my-nat-gateway2 → choose our VPC: my-vpc-01 → Create.

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
my-nat-gateway2
The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☒ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☐ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC
Select a VPC in which to create the regional NAT gateway.
vpc-00a1651d57d3137a6 (my-vpc-01)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ **Public**
☐ **Private**

Method of Elastic IP (EIP) allocation [Info](#)
Choose how IP addresses are associated with NAT gateways.

☒ **Automatic**
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling-- adding AZs automatically allocates EIPs, simplifying management.

☐ **Manual**
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q my-nat-gateway2 X [Remove](#)

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create NAT gateway](#)

Step no:8 create a security group for the Public subnet

named as:My-public-Sg1,

VPC : My-vpc-01

Inbound rules → add,

Ssh → 0.0.0.0/0

Icmp → 0.0.0.0/0

Save and create.

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

My-public-sg1

Name cannot be edited after creation.

Description [Info](#)

My-public-sg1

VPC [Info](#)

vpc-00a1651d57d3137a6 (my-vpc-01)

Inbound rules [Info](#)

Type [Info](#)

SSH

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source [Info](#)

Anywhe...

0.0.0.0/0

Description - optional [Info](#)

Delete

All ICMP - IPv4

ICMP

All

Anywhe...

0.0.0.0/0

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type [Info](#)

All traffic

Protocol [Info](#)

All

Port range [Info](#)

All

Destination [Info](#)

Custom

0.0.0.0/0

Description - optional [Info](#)

Delete

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Create security group

Once created then copy the Security group id of My-public-sg1.
(sg-08b703bbce87c0f21)

sg-08b703bbce87c0f21 - My-public-sg1

Details

Security group name

My-public-sg1

Owner

314383175192

sg-08b703bbce87c0f21

sg-08b703bbce87c0f21

Inbound rules count

2 Permission entries

Step no:9 create a security group for the Private subnet named as:My-private-security-group

VPC : My-vpc-01

Inbound rules → add,

All traffic → Paste the ID of our Public Security Group.

Save and create.

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

my-private-security-group

Name cannot be edited after creation.

Description Info

my-private-security-group

VPC Info

vpc-00a1651d57d3137a6 (my-vpc-01)

Inbound rules Info

Type Info

All traffic

Protocol Info

All

Port range Info

All

Source Info

Custom

Description - optional Info

Delete

Add rule

Outbound rules Info

Type Info

All traffic

Protocol Info

All

Port range Info

All

Destination Info

Custom

Description - optional Info

Delete

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel

Create security group

Step no:10 Go tour public route table → routes → Add 0.0.0.0/0 → internet gateway → choose the one we created.

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Add route

Cancel Preview Save changes

Step no:11 Go tour private route table → routes → Add 0.0.0.0/0 → NAT gateway → choose the one we created .

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Add route

Cancel Preview Save changes

Step no: 12 Create a public EC2 instance named as : My -public-instance,
With a keypair.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

My-Public-Instance1

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

[Quick Start](#)



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-02f986bab3de34d0d (64-bit (x86), uefi-preferred) / ami-0d62c32deead0d6c5 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260330.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86)	uefi-preferred	ami-02f986bab3de34d0d	2026-03-30	ec2-user	Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand Linux base pricing: 0.0104 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

mykeypair

[Create new key pair](#)

Network settings,

VPC : my-vpc-01

Subnet: my-public-subnet-1

Assign public IP: Enable

Select existing security group: My-public-sg1

Launch instance.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

mykeypair

Create new key pair

▼ Network settings [Info](#)

VPC - required | [Info](#)

vpc-00a1651d57d3137a6 (my-vpc-01)

Subnet | [Info](#)

subnet-0713027daa63722df my-public-subnet-1

Auto-assign public IP | [Info](#)

Enable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups | [Info](#)

Select security groups

My-public-sg1 sg-08b703bbce87c0f21

Advanced network configuration

Step no: 13 Create a private EC2 instance named as : My -public-instance,
With a keypair.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

My-private-instance2

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-02f986bab3de34d0d (64-bit (x86), uefi-preferred) / ami-0d62c32deead0d6c5 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Network settings,

VPC : my-vpc-01

Subnet: my-private-subnet-1

Assign public IP: disable

Select existing security group: My-private-sg1

Launch instance.

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

mykeypair

Create new key pair

▼ Network settings Info

VPC - required Info

vpc-00a1651d57d3137a6 (my-vpc-01)

Subnet Info

subnet-08015c59f2b49036d my-private-subnet-2

Auto-assign public IP Info

Disable

Firewall (security groups) Info

Create security group

Select existing security group

Common security groups Info

Select security groups

my-private-security-group sg-0e99911f76db158bd

Advanced network configuration

Copy the private IP of the private instance. (it will have only one address, that's also private).

Instance summary for i-086d29e5c9abab62e (My-private-instance2) Info

Updated less than a minute ago

Instance ID

IPV6 address

Hostname type

Answer private resource DNS name

Auto-assigned IP address

IAM role

IMDSv2

Operator

Public IPv4 address

Instance state

Private IP DNS name (IPv4 only)

Instance type

VPC ID

Subnet ID

Instance ARN

Private IPv4 address copied

Public DNS

Elastic IP addresses

AWS Compute Optimizer finding

Auto Scaling Group name

Managed

Step no:14 Choose our public instance → connect → ping the private ip of the private instance.

```
#  
~\#### Amazon Linux 2023  
~~\#####  
~~\###|  
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~'-'>  
~~~~  
~~.-.  
_/_/_/_/  
_/_/_/_/  
[ec2-user@ip-10-0-1-158 ~]$ ping 10.0.2.83  
PING 10.0.2.83 (10.0.2.83) 56(84) bytes of data:  
64 bytes from 10.0.2.83: icmp_seq=1 ttl=127 time=1.64 ms  
64 bytes from 10.0.2.83: icmp_seq=2 ttl=127 time=1.30 ms  
64 bytes from 10.0.2.83: icmp_seq=3 ttl=127 time=1.13 ms  
^C  
--- 10.0.2.83 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2003ms  
rtt min/avg/max/mdev = 1.131/1.359/1.644/0.213 ms  
[ec2-user@ip-10-0-1-158 ~]$
```

i-0fc04ed0018db6cf7 (My-Public-instance1)

PublicIPs: 3.143.248.63 PrivateIPs: 10.0.1.158

Step no: 15 To make ssh connection with the private instance,

Create an empty file named as **mykeyfr** and paste the content of our key pair in that.

```
[ec2-user@ip-10-0-1-158 ~]$ vi mykeyfrc
[ec2-user@ip-10-0-1-158 ~]$ cat mykeyfrc
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAtnranjIVGOT/L7XZ4BFARqR8aaI/WtXMorUyEiywAw/3uoe5
mEJ34RQMuvEMrlhYEaDBGGXKkpIxmQDXUaxx/1EbFhdVtjQeQkCypUzWjE3uKFWO
bTVNnt76K6tsgkls16wCeW0iAr9hlR5CIhQKyG8TslF6DZ/sbCMjE6TcXT1GyIAG
cLmjODRHOQxQMBUm1BSr5JisbHk2r5xQ779MNEjLKT1VzHcQnGiUyXfbF08bHf26
LdiMfXV8OmqaPDF2u3aHQwxGyDC1Z9kc4Sn3fKes/L//e26c4vtB7UvwZm2ez027
```

To make ssh connection: `ssh -i mykeyfrc ec2-user@private_ip_of_the_private_instance.` → Yes.

```
[ec2-user@ip-10-0-1-158 ~]$ ssh -i mykeyfrc ec2-user@10.0.2.83
The authenticity of host '10.0.2.83 (10.0.2.83)' can't be established.
ED25519 key fingerprint is SHA256:gfnL5/LzCn5E05F34qrSbUax4wyO2oVYXiN3E4qdzOk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.83' (ED25519) to the list of known hosts.
#####
@
#####
WARNING: UNPROTECTED PRIVATE KEY FILE!
@
#####
Permissions 0644 for 'mykeyfrc' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "mykeyfrc": bad permissions
ec2-user@10.0.2.83: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-1-158 ~]$ chmod 400 mykeyfrc
```

Permission will be denied → Change the mykeyfrc file permission using: `chmod 400 mykeyfrc`

Enter the ssh command again.

Now it will connect. To check ping any Ip address outside the instances (public and private).

Eg. ping www.google.com

It connects → Nat gateway works and the private Ec2 can be accessed via the public EC2.

```
[ec2-user@ip-10-0-1-158 ~]$ chmod 400 mykeyfrc
[ec2-user@ip-10-0-1-158 ~]$ ssh -i mykeyfrc ec2-user@10.0.2.83
#
~\  ###
~~ \#####\
~~  \###\
~~   \#/      https://aws.amazon.com/linux/amazon-linux-2023
~~    V~'-'>
~~~
~~.
~~/_/
~~/_m/'-'>
[ec2-user@ip-10-0-2-83 ~]$ ls
[ec2-user@ip-10-0-2-83 ~]$ ping www.google.com
PING www.google.com (142.251.150.119) 56(84) bytes of data.
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=1 ttl=116 time=8.58 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=2 ttl=112 time=8.26 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=3 ttl=112 time=8.24 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=4 ttl=116 time=8.30 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=5 ttl=116 time=8.28 ms
^C
--- www.google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4008ms
rtt min/avg/max/mdev = 8.242/8.330/8.575/0.123 ms
```